

OSCILLOSCOPES AVAILABLE IN THE MARKET

Manufacturer	Product/Series	Description	Model Nos.	Bandwidth	Sample rate	Record length
Keysight Technologies	Initium S-Series	10-bit ADC; low-noise front-end technology; touch capability	14 (seven DSO models and seven MSO models)	500MHz to 8GHz	20GSa/s (2 channels); 10GSa/s (4 channels)	100Mpts (2 channels); 50Mpts (4 channels)
Keysight Technologies	Initium V-Series	160-bit hardware serial trigger; 20GSa/s digital channels; standard 500GB removable solid-state drive	18 (six DSO models, six DSA models and six MSO Models)	8GHz to 33GHz	80GSa/s - 2Gpts	50Mpts - 2Gpts
LabNation	Smartscope	Open source USB oscilloscopes designed to run on multiple operating systems and platforms such as smartphones, tablets and PCs	-	30MHz -3dB point	2×100 MS/s	-
OWON	XDS series n-in-1 DSO	Integrates data logger, multimeter, waveform generator, DSO and more into one device, through its Wi-Fi module to co-share measurement data	Four (XDS3102, XD-S3102A, XDS3202, XDS3202A)	100MHz to 200MHz	1GS/s - 2GS/s	40Mpts
Rohde & Schwarz	HMO1202 mixed-signal oscilloscopes	Mixed-domain oscilloscope series ideal for use in development of embedded designs and analysing serial SPI buses on digital channels while simultaneously monitoring two analogue signals referenced to transmitted SPI data	Three (HMO1212, HMO1222, HMO1232)	100MHz/200MHz/300MHz	Up to 2GS/s	-
SIGLENT Technologies	SDS2000X series super phosphor oscilloscopes	Available in bandwidths of 70MHz to 300MHz, maximum sample rate of 2GSa/s and maximum record length of 140Mpts; most-commonly-used functions can be accessed with its user-friendly one-button design	Eight (SDS2304X, SDS2302X, SD-S2204X, SDS2202X, SDS2104X, SD-S2102X, SDS2074X, SDS2072X)	70MHz to 300MHz	Maximum 2GSa/s	140Mpts
Teledyne LeCroy	LabMaster 10-100Zi 100GHz oscilloscope	With 100GHz bandwidth and 240GS/s sampling rate, acquires the fastest signals generated by scientists and engineers	Eight (available in 20GHz, 25GHz, 30GHz, 36GHz, 50GHz, 59GHz, 65GHz, 100GHz models)	20GHz to 100GHz	240GS/s	20Mpts - 1024Mpts
Teledyne LeCroy	WaveMaster 8 Zi-A	Up to 30GHz bandwidth and 80GS/s sample rate; 14.1Gb/s serial trigger; integrated 50Ω and 1MΩ inputs for true connection and probing flexibility	Eight	4GHz to 30GHz	80GS/s	32Mpts - 512Mpts

Chivukula, technical marketing engineer, National Instruments.

Another thing is that scopes cannot show time domain because it happens too fast for humans to make sense of it. Let us say we have a waveform that is rising and falling periodically. The scope always

shows the data with respect to the trigger. Traditionally, the trigger has always been a voltage level with respect to which the waveform is shown. Over time people have started adding sophisticated triggers; there are some triggers on a digital signal where there are specific bits

that start the trigger.

Chivukula adds, "In our case, you can program a trigger. With this we have made available the ability to define a very custom trigger—a user-defined trigger limits where you can draw the trigger you want and it will be where the signal is